

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 16



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Recap of Previous Lecture



Topic

Order of an element in a group



Topic

Sub-group



Topics to be Covered



✓
Topic

Cyclic group

✓
Topic

Properties w.r.t. Cyclic group

✓
Topic

Some important properties

Cyclic Group



Topic : Cyclic group

* Let $(G, *)$ is a group. If there exists any element $a \in G$ such that Every element of set G can be represented in the form $(a)^n$ for some positive integer 'n', then

*binary opⁿ w.r.t. $(G, *)$*
Can not be '0'

$\underbrace{a * a * \dots * a}_{\text{'n times a'}}$

group $(G, *)$ is called a cyclic group. and element 'a' is one of the generator of cyclic group $(G, *)$

Note: A Cyclic group may have more than one generator

Note: Let $(G, *)$ be the group and 'e' is the identity element of the group.

✓ $(e)^1 = e$

✓ $(e)^2 = e * e = e$

✓ $(e)^3 = e^2 * e = e * e = e$

⋮

Identity element can not generate any other element except itself.

∴ Identity element can not be the generator of a group containing any other element apart from identity element

Q: $\{1, -1\}$ is a group w.r.t. multiplication

Here $\odot$ $1 = \text{identity}$ $\therefore$ can not generate any other element except itself

$\left. \begin{array}{l} (-1)^1 = -1 \\ (-1)^2 = 1 = e \end{array} \right\}$ all the element of the group
can be generated by '-1'

$O(-1) = 2 = O(G)$

$\therefore \{1, -1\}$ w.r.t. multiplication is a
cyclic group and '-1' is the
generator of the cyclic group.

eg: ① Check whether the group $\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group or not, if cyclic then find all the generators of the cyclic group $O(G)=3$

$1 = \text{identity}$ (Not a generator)
 $O(1) = 1 \neq O(G)$

$$(\omega)^1 = \omega$$
$$(\omega)^2 = \omega^2$$

$$(\omega)^3 = \omega^3 = 1 = \text{identity}$$

$$O(\omega) = 3 = O(G)$$

$\therefore \omega$ is a generator

all elements
of set

$$(\omega^2)^1 = \omega^2$$
$$(\omega^2)^2 = \omega$$

$$(\omega^2)^3 = 1 = \text{identity}$$

$$O(\omega^2) = 3 = O(G)$$

$\therefore \omega^2$ is a generator

all elements

$\{1, \omega, \omega^2\}$ w.r.t. multiplication is a cyclic group, with ω & ω^2 as generators

eg: ② Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the Cyclic group $O(G)=4$

1 = identity
(Not a generator)

$$O(1) = 1 \neq O(G)$$

$$(-1)^1 = -1$$

$$(-1)^2 = 1 = e$$

$$(-1)^3 = -1$$

$$(-1)^4 = 1$$

$$(-1)^5 = -1$$

$$(-1)^6 = 1$$

$$O(-1) = 2 \neq O(G)$$

elements
are repeated

eg: ② Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the Cyclic group $O(G)=4$

1 = identity
(Not a generator)

$$O(1) = 1 \neq O(G)$$

$$(-1)^1 = -1$$

$$(-1)^2 = 1 = e$$

$$O(-1) = 2 \neq O(G)$$

If order of an element is not Equal to the order of the group then it will generate the identity element before it has generated all the elements of the set. And now if we further increase the power of the element then it will repeat the elements that it has already generated, and can not generate remaining elements of the set
∴ It Can not be the generator of the set

eg: ② Check whether the group $\{1, -1, i, -i\}$ w.r.t. multiplication is a cyclic group or not. if cyclic then find all the generators of the Cyclic group

1 = identity
(Not a generator)
 $O(1) = 1 \neq O(G)$

$$\begin{aligned} (-1)^1 &= -1 \\ (-1)^2 &= 1 = \text{identity} \end{aligned}$$

$$\begin{aligned} (-1)^3 &= -1 \\ (-1)^4 &= 1 \end{aligned}$$

$$\begin{aligned} (-1)^5 &= -1 \\ (-1)^6 &= 1 \end{aligned}$$

$$O(-1) = 2 \neq O(G)$$

elements will be repeated

$$\begin{aligned} (i)^1 &= i \\ (i)^2 &= i^2 = -1 \\ (i)^3 &= i^2 \cdot i = -i \\ (i)^4 &= i^2 \cdot i^2 = 1 = \text{identity} \end{aligned}$$

$$O(i) = 4 = O(G)$$

$\therefore i$ is a generator

$$\begin{aligned} (-i)^1 &= -i \\ (-i)^2 &= i^2 = -1 \\ (-i)^3 &= +i \\ (-i)^4 &= 1 = \text{identity} \end{aligned}$$

$$O(-i) = 4 = O(G)$$

$\therefore -i$ is a generator

all elements

$\{1, -1, i, -i\}$ w.r.t. multiplication is a Cyclic group, with i & $-i$ as generator



Topic : Properties w.r.t. Cyclic group

① Identity element can not be the generator of the group containing any other element except itself



Topic : Properties w.r.t. Cyclic group

② Let $(G, *)$ be a group of order $= |G|$
if there exist some element $a \in G$
such that $O(a) = |G| = \text{Order of group}$
then $(G, *)$ is a cyclic group
and 'a' is one of its generator



Topic : Properties w.r.t. Cyclic group

③ If $a \in G$ is the generator of cyclic group $(G, *)$ then $\text{inv}(a) = a^{-1}$ will also be the generator of that cyclic group

We know $O(a^{-1}) = O(a)$

$$\therefore O(a^{-1}) = O(G)$$

Hence a^{-1} is also the generator



Topic : Properties w.r.t. Cyclic group

④ Let $(G, *)$ be a group of order $= |G|$
if there exist no element $a \in G$
s.t. $O(a) = |G| = \text{order of group}$
then group is not a Cyclic group

ie, if $O(a) \neq O(G), \forall a \in \text{group } (G, *)$
then $(G, *)$ is not a cyclic group

Example on next slide.

eg: ③ Check whether the group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a cyclic group or not, if cyclic then find all the generators of the cyclic group ($\phi(8)=4$) ✓

1 - identity
(Not a generator)

$$\boxed{O(1) = 1 \neq \phi(8)}$$

∴ Can not be the generator

$$(3)^1 = 3$$

$$(3)^2 = 3 \otimes_8 3 = 1 \text{ identity}$$

$$\boxed{O(3) = 2 \neq \phi(8)}$$

∴ Can not be the generator

$$(5)^1 = 5$$

$$(5)^2 = 5 \otimes_8 5 = 1 \text{ (identity)}$$

$$\boxed{O(5) = 2 \neq \phi(8)}$$

∴ Can not be the generator

$$(7)^1 = 7$$

$$(7)^2 = 7 \otimes_8 7 = 1 \text{ (identity)}$$

$$\boxed{O(7) = 2 \neq \phi(8)}$$

∴ Can not be the generator

No generator, ∴ Group $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is not a cyclic group

Q:- Let $\{0, 1, 2, 3, 4\}$ w.r.t. binary opⁿ $\oplus_5$ is a cyclic group, then find all generators of the cyclic group
order of group = 5 (Prime number)

0 = identity

∴ '0' can not be the generator

∴ Order of every non-identity element will be same as order of group 'G'

ie., $O(1) = O(2) = O(3) = O(4) = 5$

∴ 1, 2, 3 & 4 all are generator of given group.

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary opⁿ $\otimes_7$ be a Cyclic group. Find all the generator of Cyclic group.

Order of group = 6

1 = identity
 $\therefore$ 1 is not a generator

$(2)^1 = 2$
 $(2)^2 = 4$
 $(2)^3 = 1 = \text{identity}$
 $O(2) = 3 \neq O(G)$
 $\therefore$ Not a generator

$\text{inv}(2) = 4$, $\therefore O(4) = O(2) = 3 \neq O(G)$
Hence '4' is not a generator

$$(3)^1 = 3$$

$$(3)^2 = 2$$

$$(3)^3 = 6$$

$$(3)^4 = 4$$

$$(3)^5 = 5$$

$$(3)^6 = 1 = \text{identity}$$

all elements

$\therefore$ '3' is a generator

$$\text{inv}(3) = 5$$

$\therefore$ '5' is also the generator

$$(6)^1 = 6$$

$$(6)^2 = 1 = \text{identity}$$

$$O(6) = 2 \neq O(G)$$

$\therefore$ 6 is not a generator



Topic : Properties w.r.t. Cyclic group

⑤ Let $(G, *)$ be a cyclic group of order $= n$,
and $a \in G$ is one of its generator
then

$\{ (a)^m \mid m \in \mathbb{Z}^+ \text{ and } m < n \text{ and } \text{GCD}(m, n) = 1 \}$
ie. m & n are Coprime
this set is the set of all generator of
Cyclic group G .

Q. Let $\{1, 2, 3, 4, 5, 6\}$ w.r.t. binary opⁿ $\otimes_7$ be a cyclic group with element '3' as one of its generator

find all generators of the cyclic group

order of group is = 6

$\Rightarrow$ Positive integers that are less than 6 and Co-prime to '6' are 1 & 5

$\rightarrow$ We know '3' is one generator

$\therefore \{ (3)^1, (3)^5 \}$ is the set of all generators of group.

$\{ 3, 5 \}$ is the set of all generators of group.



Topic : Properties w.r.t. Cyclic group

⑤ Let $(G, *)$ be a cyclic group of order $= n$,
and $a \in G$ is one of its generators
then

$$\{ (a)^m \mid m \in \mathbb{Z}^+ \text{ and } m < n \text{ and } \text{GCD}(m, n) = 1 \}$$

is the set of all the generators of the cyclic group

⑥ Let $(G, *)$ be a cyclic group of order $= n$, then
number of generators of the cyclic group will be
same as number of positive integers that are less than 'n'
and Co-prime to 'n'

Q: Let $(G, *)$ be a cyclic group of order = 50,
find the number of generators of cyclic group $(G, *)$

→ Number of generators of the cyclic group
will be same as no. of +ve integers that
are less than 50 & Coprime to 50.

→ Such numbers are

1, 3, 7, 9, 11, 13, 17, 19, 21, 23, 27, 29, 31, 33,
37, 39, 41, 43, 47, 49

→ '20' such numbers are there $\therefore$ No. of generators = 20

★ Euler's totient function $\phi(n)$

- ★ Let 'n' is any fixed positive integer, then number of positive integers that are less than 'n' and Coprime to 'n' can be given by Euler's totient function $\phi(n)$

$$\phi(n) = n \cdot \left\{ \left(1 - \frac{1}{p_1}\right) \cdot \left(1 - \frac{1}{p_2}\right) \cdot \left(1 - \frac{1}{p_3}\right) \cdots \right\}$$

$p_1, p_2, p_3, \dots$ etc are distinct prime factors of 'n'.

Q: Let $(G, *)$ be a cyclic group of order = 50,
find the number of generators of cyclic group $(G, *)$

Order of group = 50

∴ Number of generators = $\phi(50)$

$$n = 50 = 2 \times 25$$

$$= (2)^1 \times (5)^2$$

2 & 5 are
the only two
distinct prime factors
of 50.

$$= 50 \times \left\{ \left(1 - \frac{1}{2}\right) \cdot \left(1 - \frac{1}{5}\right) \right\}$$

$$= 50 \times \frac{1}{2} \times \frac{4}{5}$$

$$= \boxed{20} \quad \underline{\underline{\text{Ans}}}$$

Q: Let $(G, *)$ be a cyclic group of order = 75
find the number of generators of cyclic group $(G, *)$

Order of group = 75

∴ Number of generators = $\phi(75)$

$$\begin{aligned} 75 &= 3 \times 25 \\ &= (3)^1 \times (5)^2 \end{aligned}$$

$$= 75 * \left\{ \left(1 - \frac{1}{3}\right) \cdot \left(1 - \frac{1}{5}\right) \right\}$$

$$= \cancel{75} * \frac{2}{\cancel{3}} * \frac{4}{\cancel{5}}$$

$$= \textcircled{40} \text{ Ans}$$

Q. Let $(G, *)$ be a cyclic group of prime order = 'p'
find the number of generators of the group

* If 'p' is a prime no., then no. of positive integers that are less than 'p' and Co-prime to p are $(p-1)$

∴ No. of generators = $(p-1)$

$$\# \text{ generators} = \phi(p) = p * \left\{1 - \frac{1}{p}\right\}$$

If 'p' is prime no.
then only prime factor
is 'p'.

$$= p * \left(\frac{p-1}{p}\right) = (p-1)$$



Topic : Some important properties

① Every group of prime order is a cyclic group.

② Every cyclic group is abelian group

Let $(G, *)$ be a group with 'a' as one of its generator
let elements $p, q \in G$ and there exists two least positive integers $m \neq n$ such that $(a)^m = p$ & $(a)^n = q$

$$\text{Now } p * q = (a)^m * (a)^n$$

$$\begin{aligned} & \text{"m times" a} \quad \quad \quad \text{"n times a"} \\ & = (a * a * \dots * a) * (a * a * \dots * a) \end{aligned}$$

$$= \overbrace{a * a * a * \dots * a}^{(m+n) \text{ times a}}$$

$$= \underbrace{(a * a * \dots * a)}_{n \text{ times a}} * \underbrace{(a * a * \dots * a)}_{m \text{ times a}}$$

$$= a^n * a^m$$

$$= q * p$$

$$p * q = q * p$$

$\therefore$ Commutative

Hence abelian group

} Because associative

③ Every group of prime order is abelian

{ By Property ① & Property ② }



Topic : Some important properties

④ Every subgroup of a cyclic group is also a cyclic group, but generators of the cyclic subgroup need not be same as generator of original cyclic group

eg. $\{1, -1, i, -i\}$ is a cyclic group with i & $-i$ as generator

↓
 $\{1, -1\}$ is a sub-group of above group, it is also cyclic, but generator is -1

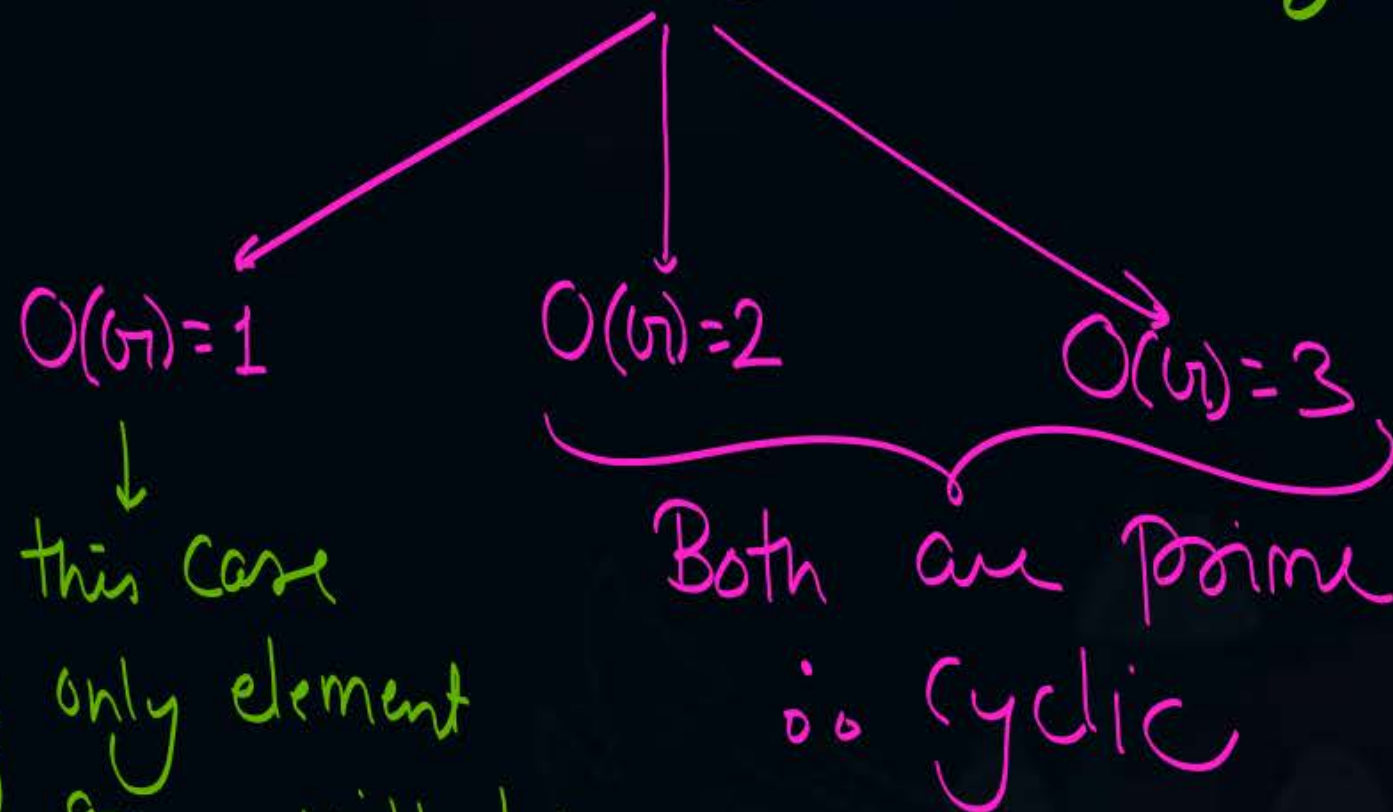
Note: If group is cyclic then its sub-group will also be cyclic.

But, if group is not cyclic, then sub-group may or may not be cyclic



Topic : Some important properties

⑤ Every group of order ≤ 3 is always cyclic



↓
in this case
the only element
of group will be
identity element,
And it can generate itself
 $\therefore$ cyclic



Topic : Some important properties

⑥ Every group of order = 4 is cyclic
Except for the groups of 'Klein-4' type

Properties of "Klein-4" group:

✓ ① Order of group = 4

g. $\{e, a, b, c\}$
identity $\swarrow$ $\nwarrow$ non-identity

✓ ② Every element is inverse of itself

i.e. $\begin{matrix} \text{inv}(e) = e & O(e) = 1 \\ \text{inv}(a) = a & O(a) = 2 \\ \text{inv}(b) = b & O(b) = 2 \\ \text{inv}(c) = c & O(c) = 2 \end{matrix}$

✓ ③ Binary opⁿ on two non-identity elements results in third non-identity element

i.e. $\begin{matrix} a * b = b * a = c \\ a * c = c * a = b \\ b * c = c * b = a \end{matrix}$

Klein-4 groups are not cyclic but they are abelian

Properties of "Klein-4" group:

✓ ① Order of group = 4

✓ ② Every element is inverse of itself

✓ ③ Binary opⁿ on two non-identity elements results in third non-identity element

identity non-identity
eg. $\{e, a, b, c\}$

i.e. $\text{inv}(e) = e$
 $\text{inv}(a) = a$
 $\text{inv}(b) = b$
 $\text{inv}(c) = c$

i.e. $a * b = b * a = c$
 $a * c = c * a = b$
 $b * c = c * b = a$

Klein-4 groups are not cyclic but they are abelian

eg. $\{1, 3, 5, 7\}$ w.r.t. $\otimes_8$ is a group of Klein-4 type

① Set = $\{1, 3, 5, 7\}$ order = 4
identity non-identity

② $\text{inv}(1) = 1$
 $\text{inv}(3) = 3$
 $\text{inv}(5) = 5$
 $\text{inv}(7) = 7$

③ $3 \otimes_8 5 = 7 = 5 \otimes_8 3$
 $3 \otimes_8 7 = 5 = 7 \otimes_8 3$
 $5 \otimes_8 7 = 3 = 7 \otimes_8 5$



Topic : Some important properties

- ⑦ 'Klein-4' group of order = 4 is not cyclic but it is an abelian group
- ⑧ Every group of order ≤ 5 is cyclic except for 'Klein-4' group of order = 4
- ⑨ Every group of order ≤ 5 is always abelian group

Q:- $Z_n = \{0, 1, 2, \dots, n-1\}$ is a group w.r.t.
binary opⁿ $\oplus_n$.

• In the group $Z_2 \times Z_3$ how many elements
are their own inverse.

$$Z_2 = \{0, 1\}$$

$$Z_3 = \{0, 1, 2\}$$

$$\mathbb{Z}_2 = \{0, 1\}$$

$$\mathbb{Z}_3 = \{0, 1, 2\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \left\{ \begin{array}{l} (0, 0), (0, 1), (0, 2) \\ (1, 0), (1, 1), (1, 2) \end{array} \right\}$$

$$\mathbb{Z}_2 \times \mathbb{Z}_3 = \{ (a, b) \mid a \in \mathbb{Z}_2 \text{ and } b \in \mathbb{Z}_3 \}$$

If $(p, q), (r, s) \in \mathbb{Z}_2 \times \mathbb{Z}_3$

then $(p, q) * (r, s) = (p \oplus_2 r, q \oplus_3 s)$

Binary opⁿ wrt.
group $\mathbb{Z}_2 \times \mathbb{Z}_3$

opⁿ wrt. $\mathbb{Z}_2$ opⁿ wrt. $\mathbb{Z}_3$



2 mins Summary



✓
Topic

Cyclic group

✓
Topic

Properties w.r.t. Cyclic group

✓
Topic

Some important properties

THANK - YOU